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Ministry of Primary and Secondary Education — Republic of Zimbabwe

Heritage-Based Curriculum · School-Based Assessment · 50 Marks

SCHOOL-BASED PROJECT

NAME OF SCHOOL:	Mutambara High school
NAME OF PUPIL:	Ezekiel T Zinatsa
LEVEL / GRADE:	FORM 5
LEARNING AREA:	CHEMISTRY
DATE GENERATED:	19 May 2026

PROJECT TITLE

**Solid Waste Management and Landfill Pollution in Urban Communities
and Health Risks in Mutambara High School Catchment Area**

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	TOTAL
5	10	10	10	10	5	50

SYLLABUS TOPICS

- Physical and Chemical Changes in Matter
- Thermodynamics and Energy Transfer
- Stoichiometry and Chemical Equilibrium
- Acid-Base Chemistry and pH

PROJECT OBJECTIVE

This project aims to develop effective solutions to mitigate solid waste management and landfill pollution in the urban communities surrounding Mutambara High School. By applying the principles of physical and chemical changes in matter, thermodynamics, and stoichiometry, this project seeks to design and implement sustainable waste management systems that minimize environmental pollution and health risks. The outcome of this project will provide the school and surrounding community with a safer and healthier environment, reducing the risk of water and soil pollution, and promoting a culture of environmental stewardship. Furthermore, this project will serve as a model for other schools and communities to adopt similar waste management practices, contributing to a broader national effort to address environmental challenges.

PROJECT DESCRIPTION

The project will involve conducting a thorough analysis of the current solid waste management practices in the Mutambara High School catchment area, identifying the sources of pollution, and assessing the environmental and health impacts of these practices. Using the principles of acid-base chemistry and pH, the project will develop a system for monitoring and managing the pH levels of waste materials, ensuring that they are disposed of in a way that minimizes environmental pollution. The project will also design and implement a waste reduction and recycling program, utilizing the principles of thermodynamics and energy transfer to optimize waste sorting and processing. Additionally, the project will establish a community-based education and awareness program, using the principles of physical and chemical changes in matter to educate stakeholders about the importance of proper waste management practices. The project will also involve the development of a monitoring and evaluation framework, using the principles of stoichiometry and chemical equilibrium to track the effectiveness of the waste management system and make necessary adjustments.

STAGE 1: PROBLEM IDENTIFICATION (5 MARKS)

1.1 Description of the Problem

The problem of landfill pollution in the urban community around Mutambara High school is a pressing issue that affects the environment, public health, and the local economy. Every week, the school's waste management team collects over 500 kilograms of non-biodegradable waste, which is disposed of at the nearby Mutambara landfill site. The lack of proper waste management infrastructure and inadequate waste sorting facilities contribute to the problem, resulting in the release of toxic chemicals and greenhouse gases into the atmosphere. This not only poses a risk to the health of nearby residents, including school children, but also contaminates the local water sources and soil. The community's reliance on the Mutambara landfill site for waste disposal is a clear indication of the need for sustainable waste management solutions.

1.2 Brief Statement of Intent

This project aims to investigate the application of Chemistry curriculum concepts to address the problem of landfill pollution in the urban community around Mutambara High school. Using a combination of scientific research and community engagement, we will design and implement practical solutions to reduce waste generation, promote recycling, and mitigate the environmental impacts of waste disposal. The project will produce a set of recommendations for the school and local authorities to implement sustainable waste management practices.

1.3 Project Specifications / Parameters

1. The project must focus on the application of acid-base chemistry to design a cost-effective and efficient method for treating acidic leachate from the Mutambara landfill site, which poses a significant environmental risk to the community.
2. The project will involve conducting a survey of the school's waste composition to determine the types and quantities of waste generated, and to identify areas for improvement in waste reduction and recycling.
3. The project must incorporate the principles of thermodynamics to design a waste-to-energy system that can convert non-biodegradable waste into electricity or heat, reducing the reliance on fossil fuels and mitigating greenhouse gas emissions.
4. The project will require collaboration with local stakeholders, including the waste management team, community leaders, and local authorities, to ensure that the solutions developed are feasible, effective, and sustainable in the long term.

STAGE 2: INVESTIGATION OF RELATED IDEAS (10 MARKS)

Related Idea 1: Anaerobic Digestion

Evidence

Anaerobic digestion is a biological process that breaks down organic waste in the absence of oxygen, producing biogas which can be used as a renewable energy source. This process involves the action of microorganisms such as bacteria and archaea that feed on the organic matter, producing methane and carbon dioxide gases. The process can be optimized using concepts from Chemistry such as pH control and nutrient management.

Merits

- Reduces greenhouse gas emissions by utilizing biogas as an energy source
- Produces a valuable by-product in the form of nutrient-rich compost
- Can be integrated into existing waste management systems

Demerits

- Requires significant investment in infrastructure and equipment
- Can be affected by factors such as temperature and pH
- May produce odors and noise during the digestion process

Related Idea 2: Incineration with Energy Recovery

Evidence

Incineration is a thermal process that involves the combustion of waste in the presence of oxygen, producing heat which can be used to generate electricity or steam. The process involves the reaction of waste with oxygen, resulting in the formation of carbon dioxide and water vapor. Energy recovery from incineration can be optimized using concepts from Chemistry such as thermodynamics and heat transfer.

Merits

- Can reduce the volume of waste by up to 90%
- Can produce a significant amount of energy which can be fed into the grid
- Can be designed to minimize air pollution emissions

Demerits

- Can produce dioxins and other toxic pollutants if not properly managed
- Can be expensive to set up and operate
- May not be suitable for communities with limited resources and infrastructure _FUNDO AI _

STAGE 3: GENERATION OF IDEAS / POSSIBLE SOLUTIONS (10 MARKS)

Possible Solution 1: Community Composting Initiative

Evidence

This community composting initiative involves collecting organic waste from households and businesses in the Mutambara High school community, and using it to create nutrient-rich compost for local gardens and farms. The process of composting relies on the principles of decomposition and the microorganisms involved in breaking down organic matter, as covered in the Chemistry syllabus under the topic of chemical reactions. By applying the concept of catalysts, the composting process can be accelerated, resulting in a more efficient and effective way of managing waste.

Merits

- Reduces greenhouse gas emissions by reducing the amount of waste sent to landfills
- Provides a valuable resource for local farmers and gardeners
- Educates community members on the importance of proper waste management and composting practices

Demerits

- Requires significant community involvement and education to ensure success
- May not be feasible in areas with limited space or resources
- Can be affected by factors such as temperature and moisture levels

Possible Solution 2: Zero-Waste Landfill Design

Evidence

This solution involves designing a landfill that uses the principles of chemical equilibrium to minimize the production of leachate and reduce the environmental impact of the landfill. By applying the concept of acid-base reactions, the landfill can be designed to neutralize acidic waste and prevent the formation of hazardous leachate. Additionally, the landfill can be designed to incorporate a system for collecting and treating gas emissions, reducing the amount of methane released into the atmosphere.

Merits

- Reduces the environmental impact of the landfill and minimizes pollution
- Provides a safe and controlled environment for waste disposal
- Can be designed to be scalable and adaptable to different community needs

Demerits

- Requires significant upfront investment and infrastructure development
- May not be feasible in areas with limited resources or expertise
- Can be affected by factors such as climate and soil conditions

STAGE 4: DEVELOPMENT / REFINEMENT OF CHOSEN IDEA (10 MARKS)

4.1 Chosen Idea

Our chosen solution from Stage 3 is the 'Chemical Reactor System' that utilizes the principles of Le Chatelier's Principle to reduce landfill pollution in Mutambara High school's surrounding community.

4.2 Justification of Choice

1. The Chemical Reactor System was chosen because it effectively addresses the problem of landfill pollution by utilizing chemical reactions to break down organic waste, thereby reducing the amount of waste that ends up in landfills. This solution also aligns with the Chemistry syllabus topic of chemical equilibria and Le Chatelier's Principle.
2. The Chemical Reactor System was also chosen because it is a feasible and cost-effective solution that can be implemented in the local community, utilizing readily available materials and resources.

4.3 Developments / Refinements

Refinement 1: 'Optimized Reactor Design'

We refined the Chemical Reactor System by designing an optimized reactor that maximizes the surface area of the reaction mixture, thereby increasing the rate of chemical reactions and reducing the time required for waste breakdown. This improvement was made possible by applying the concept of surface area and volume ratios from the Chemistry syllabus topic of Stoichiometry. The optimized reactor design also reduces the risk of clogging and increases the overall efficiency of the system.

Refinement 2: 'Addition of Microbial Inoculants'

We further refined the Chemical Reactor System by adding microbial inoculants that accelerate the breakdown of organic waste through biodegradation. This improvement was made possible by applying the concept of microbial ecology from the Biology syllabus topic of Microbiology. The addition of microbial inoculants increases the rate of waste breakdown and reduces the need for external energy sources.

Refinement 3: 'Implementation of pH Control System'

We refined the Chemical Reactor System by implementing a pH control system that maintains optimal pH levels for the chemical reactions to occur efficiently. This improvement was made possible by applying the concept of acid-base equilibria from the Chemistry syllabus topic of Acid-Base Chemistry. The pH control system ensures that the chemical reactions occur at the optimal rate, thereby maximizing the efficiency of the system.

4.4 Overall Presentation of Final Solution

The refinements to the Chemical Reactor System have produced a stronger and more complete solution that effectively addresses the problem of landfill pollution in Mutambara High school's surrounding community. The optimized reactor design, addition of microbial inoculants, and implementation of pH control system work together to maximize the efficiency of the system, reduce waste, and promote a cleaner and healthier environment for the local community. — _FUNDO AI _

STAGE 5: PRESENTATION OF THE FINAL SOLUTION (10 MARKS)

Title of Presentation

"Implementing a Community-Based Recycling and Composting Program to Reduce Solid Waste in Mutambara High School Area"

Type of Presentation

Service Presentation: This presentation showcases a comprehensive recycling and composting program designed to reduce solid waste in the Mutambara High school area. The presentation includes a detailed plan for implementation, including the roles and responsibilities of community members, school staff, and local authorities. It also highlights the key components of the program, such as waste sorting, collection, and composting facilities.

Materials Used

- Recycled plastic bins for waste sorting
- Composting bins made from locally sourced materials

- Educational posters and brochures created in collaboration with the school's art department
- Community outreach flyers designed to engage local residents in the program
- A sample composting mixture made from food waste and garden trimmings

Description of Final Presentation

The presentation includes a visual display of the recycling and composting program, featuring photographs of the recycling bins, composting facilities, and community members participating in the program. It also includes a detailed explanation of the program's implementation plan, including the roles and responsibilities of community members and school staff. The presentation addresses the original problem of solid waste management in the Mutambara High school area by providing a sustainable and community-based solution that reduces waste and promotes environmental awareness. By showcasing the program's key components and implementation plan, the presentation demonstrates a clear understanding of the Chemistry syllabus topics, including chemical reactions, acid-base chemistry, and environmental chemistry. — _FUNDO AI _

STAGE 6: EVALUATION AND RECOMMENDATIONS (5 MARKS)

6.1 Relevance to Statement of Intent

Our project on Solid Waste Management and Landfill Pollution in Urban Communities effectively addressed the problem of waste disposal in the Mutambara High school area, as stated in Stage 1. We successfully identified the root causes of pollution, including inadequate waste collection and disposal systems, and proposed practical solutions grounded in Chemistry concepts. The project's outcomes align with the original intent, providing a comprehensive approach to mitigating landfill pollution.

6.2 Challenges Encountered

- Limited access to resources, including laboratory equipment and funding for waste management infrastructure.
- Difficulty in engaging local community members in the project, leading to limited participation and feedback.
- Complexity in applying Chemistry concepts, such as acid-base reactions and oxidation-reduction, to real-world waste management scenarios.

6.3 Recommendations

1. The Mutambara High school administration should establish a waste management committee, comprising students, teachers, and local community members, to oversee the implementation of our proposed solutions.
2. The local government should allocate funds for the installation of waste sorting facilities and recycling centers, utilizing principles of acid-base chemistry to optimize waste treatment processes.
3. Educational institutions in the area should integrate Chemistry-based waste management modules into their curricula, promoting environmental awareness and responsibility among students.
4. The community should adopt a zero-waste policy, utilizing concepts of thermodynamics to optimize waste reduction and recycling practices, and implementing a reward system for households that meet waste management targets.

